OPI LAB · DH2 · 2026-10-23

OPI NVIDIA DPF Adapter

Bugs, issues, errors, and their fixes — in detail

Every schema and API defect found while pinning, translating, and running the adapter against the live k3s cluster on dh2. Each item records the error, the root cause, the code or cluster change, the files, and the lab proof.

Repo	opi-nvidia-dpf-adapter
Lab	dh2 172.22.1.2 — NVIDIA BlueField-2 — k3s v1.36.4+k3s1
DPF	nvcr.io/nvidia/doca/dpf-system:v26.4.0 window [v26.4.0, v27.0.0)
OPI	config.openshift.io/v1 @ c107ef71 unmerged PR #6 (Helm chart)
K8s modules	k8s.io/* v0.36.0
Adapter ns	opi-nvidia-dpf-adapter-system
Final lab run	All mapped APIs LIVE. Dead APIs: NONE.

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1. Summary of every issue

Thirteen defects blocked enablement, translation, or safe multi-vendor behavior. The table is the index. Sections 3.1–3.13 are the full write-ups.

1	DPF window v25 vs lab v26.4.0	VendorOperatorSupported would be TooNew	Pin window [v26.4.0, v27.0.0)	Supported=True
2	No version pin file	Fixtures, go.mod, and docs could drift	config/versions.yaml + lock tests	verify-api PASS
3	SFC dropped Helm chart name	DPUService had repo+version, no chart	Write spec.helmChart.source.chart	hbn chart set
4	Chart fields treated as upstream-main	Silent drop if checkout moves to main	Pin PR #6 unmerged + fixture CEL	sfc-chart-probe admits
5	CEL ref.Val blocked SSA	json: unsupported type map[ref.Val]ref.Val	celNative unwrap in pkg/mapping/cel.go	hbn-ifc + chain live
6	DPUDeployment would own children	Two controllers on same DPUService/BFB	Direct emit only Service/Ifc/Chain	3 children only
7	Dead write spec.security.privileged	.spec.security: field not declared; zero children	Omit-on-unknown + replace v26.4 CRDs; fix RBAC/CM	privileged=true, children exist
8	spec.vendor unknown field	Dry-run unknown field; annotation-only enablement	Enum nvidia|amd + SanitizeVendorToken	vendor=nvidia live
9	SFC status.conditions pruned	Empty status object; DPFReady dropped	Add conditions; schema: conditions	DPFReady=True
10	AMD stub CRDs on BlueField host	Fake DSC* API; empty serial allowed	Serial gate, harden stubs, delete from dh2	No AMD CRDs
11	LCM no-op / Ready collision	spec.vendor pruned; Ready would flap printer column	Annotation fallback; VendorOperatorSupported / DPFReady	Conditions owned correctly
12	Missing ClusterRole verbs	Forbidden list interfaces/chains	Apply updated manager ClusterRole	No Forbidden
13	Stale mappings ConfigMap	CrashLoop: status.schema is required	Refresh ConfigMap from shipped YAML	Pods Running

2. Pins and API families

The adapter owns neither OPI nor DPF APIs. It watches unstructured OPI objects and server-side-applies unstructured DPF objects. Kind-specific logic is YAML in config/mappings/. Pins are config/versions.yaml. The runtime window is pkg/lifecycle/version.go.

OPI	config.openshift.io/v1 — DpuOperatorConfig, DataProcessingUnit, ServiceFunctionChain
DPF provisioning	provisioning.dpu.nvidia.com/v1alpha1 — DPUDevice, DPUFlavor, BFB, DPU
DPF services	svc.dpu.nvidia.com/v1alpha1 — DPUService, DPUServiceInterface, DPUServiceChain
AMD stubs	dpu.amd.com/v1alpha1 — envtest only; deleted from dh2
Never emit	DPUDeployment, DPUServiceTemplate, DPUServiceConfiguration
Status writers	Ready = daemon. DPFReady / DSCReady = mappings. VendorOperatorSupported = LCM.

3. Issues in detail

3.1. DPF support window said v25; lab runs v26.4.0

WHAT BROKE / ERROR

- Lifecycle Manager window was [v25.1.0, v26.0.0).
- dh2 runs nvcr.io/nvidia/doca/dpf-system:v26.4.0 (Helm chart dpf-operator-v26.4.0).
- ClassifySkew would have marked that install TooNew and published VendorOperatorSupported=False.

ROOT CAUSE

- The window was written before the fixtures and the running operator were compared. It was an aspiration, not a claim.

HOW IT WAS FIXED

- Set supportedVersions["nvidia"] to MinInclusive v26.4.0, MaxExclusive v27.0.0.
- Copied the same window into config/versions.yaml (dpf.release = v26.4.0).
- Updated tests so v26.4.0 is Supported and v25.4.0 is TooOld.

FILES

- pkg/lifecycle/version.go
- config/versions.yaml
- pkg/lifecycle/version_test.go, lifecycle_controller_test.go, test/conformance/dimensions_test.go

HOW WE VERIFIED

- ClassifySkew(nvidia, v26.4.0) = Supported.
- Live dpu-operator-config: VendorOperatorSupported=True reason=Supported, message names [v26.4.0, v27.0.0).

3.2. No pin file — fixtures, go.mod, and docs could drift

WHAT BROKE / ERROR

- Kubernetes minor, OPI commit, DPF window, and go.mod lived in comments and separate tests.
- The v25 window survived a v26.4 lab install because nothing failed when they disagreed.

ROOT CAUSE

- There was no single machine-checkable pin.

HOW IT WAS FIXED

- Added config/versions.yaml (K8s v1.36.4 / go.mod v0.36.0, OPI c107ef71 PR #6 unmerged, DPF [v26.4.0, v27.0.0)).
- Lock tests: TestPinnedVersionsAgree, TestPinnedNVIDIAWindowMatchesVersionsYAML, TestHybridDesignDocIsTheContract.
- Design contract docs/hybrid-integration-design.md. make verify-api is the gate.

FILES

- config/versions.yaml
- pkg/mapping/crd_agreement_test.go
- pkg/mapping/hybrid_design_test.go
- docs/hybrid-integration-design.md

HOW WE VERIFIED

- make verify-api PASS on 2026-10-23.

3.3. ServiceFunctionChain mapping dropped the Helm chart name

WHAT BROKE / ERROR

- OPI requires networkFunctions[].chart.name.
- The mapping copied repository and version only.
- Helm would receive a repo URL and a version with no chart identity.

ROOT CAUSE

- A walk of every mapping to:/from: against vendored CRDs showed the missing path.

HOW IT WAS FIXED

- Added to config/mappings/servicefunctionchain.yaml: to spec.helmChart.source.chart, CEL item.chart.name, required: true.
- Same field on test/conformance/testdata/servicefunctionchain-xns.yaml.
- Tests fail if the chart name is missing (translation, conformance, e2e).

FILES

- config/mappings/servicefunctionchain.yaml

HOW WE VERIFIED

- Live DPUService/hbn: spec.helmChart.source.chart=hbn.
- TestMappingFieldsExistInVendoredCRDs PASS.

3.4. Chart fields treated as if they were on upstream main

WHAT BROKE / ERROR

- spec.networkFunctions[].chart exists only on unmerged opiproject/dpu-operator#6.
- A silent move to upstream main would drop chart and stop emitting children with no compile error.

ROOT CAUSE

- Chart was copied into fixtures without recording that it is a patch.

HOW IT WAS FIXED

- Pin opi.patchState: unmerged, pullRequest #6, commit c107ef71.
- Fixtures include chart plus CEL has(self.image) != has(self.chart).
- TestSFCFixturesMatchPinnedHelmChartPatch fails if those disappear.
- Mapping when: 'chart' in item so image-only NFs emit nothing.

FILES

- config/versions.yaml
- test/e2e/crds/opi/config.openshift.io_servicefunctionchains.yaml
- pkg/mapping/crd_agreement_test.go

HOW WE VERIFIED

- Live default/sfc-chart-probe admits chart hbn v25.10.1.
- Image-only NF creates no DPUService.

3.5. CEL ref.Val wrappers blocked Server-Side Apply

WHAT BROKE / ERROR

- Chain switches and interface labels are CEL lists/maps.
- cel-go returned []ref.Val and map[ref.Val]ref.Val.
- Error: apply DPUServiceInterface/hbn-ifc: failed to marshal apply configuration: json: unsupported type: map[ref.Val]ref.Val.
- ConvertToNative(interface{}) was not enough; nested maps/lists stayed as CEL types.

ROOT CAUSE

- The interpreter surface was not unwrapped to JSON-native Go types before unstructured SSA.

HOW IT WAS FIXED

- pkg/mapping/cel.go celNative recursively unwraps ref.Val, []ref.Val, and map[ref.Val]ref.Val into []any / map[string]any / string / bool / int64 / float64.
- evalCEL returns that native value. Mapping stays CEL; no Go switch on Kind.

FILES

- pkg/mapping/cel.go

HOW WE VERIFIED

- TestSFCApplyEmitsOwnedGraph: label is the Go string hbn; one switch.
- Live hbn-ifc label service=hbn; chain ports matchLabels.service=hbn.

3.6. Direct SFC ownership — never emit DPUDeployment

WHAT BROKE / ERROR

- Emitting DPUDeployment would let DPF's controller create the same DPUService, interface, and chain.
- It would also create DPUSets against BFB/Flavor already owned by the DataProcessingUnit mapping.
- Two controllers on one child is forbidden.

ROOT CAUSE

- A DPUDeployment-led graph looks convenient and is the DPF happy path, but it collides with the DPU mapping.

HOW IT WAS FIXED

- Mapping emits only DPUService (item.name), DPUServiceInterface (item.name-ifc), DPUServiceChain (source name).
- Owner is the OPI ServiceFunctionChain. Same-ns: controller ownerRef. Cross-ns: annotation + finalizer translation.opi.nvidia.com/cleanup.
- Tests: TestSFCMappingIsDirectOwnership, TestNoTwoMappingsOwnTheSameDPFKind.

FILES

- config/mappings/servicefunctionchain.yaml
- pkg/mapping/ownership_test.go

HOW WE VERIFIED

- Tests PASS. Live children of sfc-chart-probe are exactly hbn, hbn-ifc, sfc-chart-probe.

3.7. Dead write: DPUService.spec.security.privileged

WHAT BROKE / ERROR

- v26.4 CEL requires security.privileged when deployInCluster is false (the default).
- Live DPUService CRD (created 2026-09-07) had no spec.security.
- Helm --take-ownership left helm.sh/resource-policy=keep, so v26.4 chart CRDs were not installed.
- Repeating translator error every ~16m: apply DPUService/hbn: failed to create typed patch object: .spec.security: field not declared in schema.
- default/sfc-chart-probe existed with a chart NF and zero children.

ROOT CAUSE

- Operator image was already v26.4.0. Provisioning/svc CRDs were still the Sept 7 schemas. DPUServiceInterface CRD was newer (2026-10-20). Mapping correctly stayed on the fixtures.

HOW IT WAS FIXED

- Kept privileged in the shipped mapping (Kind/conformance use v26.4 CEL).
- pkg/translation/dead_fields.go: read live CRD OpenAPI, omit undeclared spec fields, retry on unknown field.
- User authorized replace of live DPUService, DPU, and DPUDevice CRDs with v26.4 fixtures; Helm keep annotations restored.
- ClusterRole lacked dpuserviceinterfaces/dpuservicechains (Forbidden list). Applied config/rbac/role.yaml as opi-nvidia-dpf-adapter-manager-role.
- New binary required status.schema on mappings; stale ConfigMap CrashLoop. Refreshed ConfigMap.

FILES

- pkg/translation/dead_fields.go
- pkg/translation/dead_fields_test.go
- pkg/translation/translation_controller.go (schema cache + prune)
- config/rbac/role.yaml
- config/mappings/servicefunctionchain.yaml

HOW WE VERIFIED

- Server dry-run with spec.security.privileged: true succeeds.
- Live DPUService/hbn has privileged=true, helmChart hbn v25.10.1.
- Children exist: hbn, hbn-ifc, sfc-chart-probe.

3.8. DpuOperatorConfig.spec.vendor was a dead field (and an injection hole)

WHAT BROKE / ERROR

- OPI CRD spec had only logLevel.
- Dry-run: unknown field spec.vendor (strict decoding).
- Enablement was annotation-only. A free-form annotation is a Helm/OLM injection hole if the LCM later execs it (path, URL, shell).

ROOT CAUSE

- The field was never on the OPI CRD. This is not DPF skew.

HOW IT WAS FIXED

- Pinned fixture + live CRD: spec.vendor enum nvidia|amd, maxLength 16, CEL self == 'nvidia' || self == 'amd'. No preserve-unknown-fields.
- Annotation lifecycle.opi.nvidia.com/vendor kept as fallback.
- SanitizeVendorToken: trim, lowercase, [a-z][a-z0-9-]{0,15}. Invalid tokens become "invalid" so KnownVendor rejects them without echoing the raw string.
- Sample sets both spec.vendor: nvidia and the annotation.

- Conformance envtest now loads pinned fixtures (sibling dpu-operator checkout still lacks the field).

FILES

- test/e2e/crds/opi/config.openshift.io_dpuoperatorconfigs.yaml
- pkg/lifecycle/version.go (SanitizeVendorToken)
- pkg/lifecycle/lifecycle_controller.go (ResolveVendor)
- config/samples/dpuoperatorconfig-nvidia.yaml
- test/conformance/suite_test.go, dimensions_test.go

HOW WE VERIFIED

- Live patch spec.vendor=nvidia accepted.
- Dry-run acme / ../../../etc/passwd / nvidia;id rejected by enum.
- Annotation acme → VendorOperatorSupported=False reason=UnknownVendor.
- TestSanitizeVendorTokenRejectsInjection PASS.

3.9. SFC status.conditions was an empty object (pruned)

WHAT BROKE / ERROR

- OPI ServiceFunctionChain status was {type: object} with no properties.
- Apiserver pruned any status write. Dry-run status patch: unknown field status.conditions.
- Mapping correctly used status.schema: pruned so the controller skipped Status().Update instead of silently dropping DPFReady.

ROOT CAUSE

- OPI-owned gap, not DPF. Upstream main still has the empty status.

HOW IT WAS FIXED

- Fixture + live CRD: standard metav1 status.conditions (required lastTransitionTime, message, reason, status, type). No preserve-unknown-fields.
- Mapping and config/condition-writers.yaml flipped to schema: conditions.
- Translator writes DPFReady only. Ready stays reserved for the daemon.

FILES

- test/e2e/crds/opi/config.openshift.io_servicefunctionchains.yaml
- config/mappings/servicefunctionchain.yaml
- config/condition-writers.yaml
- pkg/translation/servicefunctionchain_controller_test.go

HOW WE VERIFIED

- Live sfc-chart-probe: status.conditions[DPFReady]=True reason=AllServicesReady.
- TestWriterTableSchemaAgreesWithVendoredCRDs PASS.

3.10. AMD stub CRDs on a BlueField host (fake-device API)

WHAT BROKE / ERROR

- Stub CRDs dscdevices/dscprofiles/dscfirmwares/dscnodepolicies.dpu.amd.com were installed on dh2 2026-10-20.
- No Pensando card on dh2 (that card is on dh1, no Kubernetes).
- deviceSerial was required but "" satisfied required (no minLength) — a fake DSCDevice could be applied.
- Firmware URL accepted file://. Loading amd-dsc200.yaml after CRD delete produced controller-runtime ERROR: no matches for kind DSCDevice.

ROOT CAUSE

- Stubs exist so envtest can prove a second vendor is data-only. Installing them on a BlueField host is a fake hardware API.

HOW IT WAS FIXED

- Mapping when requires a non-empty dpu.amd.com/serial-number on every emit and on status (otherwise DSCReady would land on a card with no serial).
- Stub CRD: serial minLength 1 + pattern; pciAddress pattern; firmware URL ^https:// only; driver enum ionic; interface enum oob|inband.
- Deleted the four stub CRDs from dh2.
- Live ConfigMap is NVIDIA-only (dataprocessingunit.yaml, servicefunctionchain.yaml). Stubs remain under test/conformance/crds/amd for envtest.

FILES

- config/mappings/amd-dsc200.yaml
- test/conformance/crds/amd/dpu.amd.com_dscdevices.yaml
- test/conformance/crds/amd/dpu.amd.com_dscfirmwares.yaml
- pkg/mapping/engine_test.go TestAMDMappingRequiresSerial

HOW WE VERIFIED

- k3s kubectl get crd | grep amd/dsc → none.
- TestAMDMappingRequiresSerial: Pensando product, no serial → 0 objects, no status.
- After dropping the mapping from the live ConfigMap: no AMD watch errors.

3.11. LCM no-op and Ready condition collision

WHAT BROKE / ERROR

- First real-apiserver run: code read spec.vendor. CRD had no field. Apiserver deleted it. vendor was always empty, reconcile stopped, no skew was ever reported. A fake client keeps the field and stayed green.
- Writing Ready on DpuOperatorConfig or DataProcessingUnit would fight the dpu-operator daemon. Ready is the printer column.

ROOT CAUSE

- Fake client vs real apiserver prune. Two writers on one condition type.

HOW IT WAS FIXED

- Annotation fallback; spec.vendor still wins when present. Both sanitized.
- LCM writes only VendorOperatorSupported.
- Mappings write only DPFReady / DSCReady.
- config/condition-writers.yaml is the lock. Mapping Validate rejects Ready.

FILES

- pkg/lifecycle/lifecycle_controller.go
- config/condition-writers.yaml
- docs/hybrid-integration-design.md

HOW WE VERIFIED

- Live config has VendorOperatorSupported, not a translator Ready.
- OPI DPU has DPFReady, not a translator Ready.

3.12. ClusterRole missing DPUServiceInterface and DPUServiceChain

WHAT BROKE / ERROR

- Live ClusterRole only listed dpuservices.
- After SFC started emitting interfaces and chains: Forbidden list dpuserviceinterfaces / dpuservicechains.

- Reconcile could not see children it had just applied.

ROOT CAUSE

- RBAC generated from mappings was not applied to the live ClusterRole name (prefix opi-nvidia-dpf-adapter-).

HOW IT WAS FIXED

- Applied updated config/rbac/role.yaml as ClusterRole opi-nvidia-dpf-adapter-manager-role.
- Also added apiextensions.k8s.io customresourcedefinitions get/list so the translator can read live OpenAPI for omit-unknown.

FILES

- config/rbac/role.yaml
- pkg/translation/zz_generated_rbac.go

HOW WE VERIFIED

- make verify-rbac PASS. No Forbidden in controller logs.

3.13. Stale mappings ConfigMap crashed the new binary

WHAT BROKE / ERROR

- After image rebuild the manager CrashLooped.
- Error: amd-dsc200.yaml: status.schema is required (and later SFC needed schema: conditions).
- The live ConfigMap was from 2026-10-20 and did not match shipped YAML.

ROOT CAUSE

- ConfigMap is a volume mount. Image change does not refresh mapping documents.

HOW IT WAS FIXED

- Recreated ConfigMap from shipped mapping files.
- On dh2, load only dataprocessingunit.yaml and servicefunctionchain.yaml (no AMD mapping on a BlueField host).

FILES

- config/mappings/*.yaml

HOW WE VERIFIED

- Pods Running. ConfigMap keys: dataprocessingunit.yaml, servicefunctionchain.yaml. Both schema=conditions.

4. Symptoms that are not dead APIs

These look like failures on kubectl get and must not be "fixed" by inventing fields or upgrading DPF from this repo.

DPU and BFB phase=Initializing, empty status.conditions	Every mapped spec field is on the CRD and on the object (serial, bfb, nodeEffect, url). The DPF operator has not finished provisioning the BlueField. Adapter has nothing left to write.
DPUService / interface / chain READY column empty	DPF has not published Ready conditions. SFC DPFReady treats empty child conditions as non-blocking; AllServicesReady is True because children exist and none report Ready=False.
DPU.spec.astraEnabled unused	Present on the v26.4 CRD after replace. Mapping does not write it. OPI has no equivalent.
DPUDevice.spec.cluster / bmcCredentialSecretName unused	Same: present, not written. Not a dead write.
No AMD DSC* objects on dh2	Correct. No Pensando card. Stub CRDs removed. Mapping not loaded.
VPN drop / ping 100% loss	F5 Access Disconnected (F5ChannelError). scutil --nc start "F5 Access". Not an API defect.
SSH: Server accepts key, we did not send a packet	Stale IdentityAgent launchd socket. Use IdentityAgent=none IdentitiesOnly=yes IdentityFile=$HOME/.ssh/opi_lab_key UseKeychain=yes KexAlgorithms=curve25519-sha256.

5. Final live lab results (2026-10-23)

VPN Connected. ping 172.22.1.2 0% loss. Adapter controller and VSP Running. No adapter ERROR / unknown field / Forbidden.

Object	Written fields / condition	Result
dpu-operator-config	spec.vendor=nvidia; VendorOperatorSupported	True / Supported
DataProcessingUnit dh2-bluefield-2	product BlueField-2, node dh2, DPFReady	False (children Initializing)
SFC default/sfc-chart-probe	chart hbn v25.10.1; DPFReady	True / AllServicesReady
DPUDevice ...-device	spec.serialNumber	MT2321XZ0KVX
DPUFlavor dpf-default-flavor	spec.dpuMode	dpu
BFB bf-bundle	spec.url (mellanox bf-bundle-2.7.0-33)	present; phase Initializing
DPU dh2-bluefield-2	dpuNodeName, dpuDeviceName, dpuFlavor, bfb, serialNumber, nodeEffect.noEffect	all present; phase Initializing
DPUService/hbn	helmChart + security.privileged	hbn v25.10.1; privileged=true
DPUServiceInterface/hbn-ifc	interfaceType / serviceID / interfaceName / network	service / hbn / net0 / default
DPUServiceChain/sfc-chart-probe	switches[0].ports.matchLabels.service	hbn
AMD DSC* CRDs	must be absent on this host	absent

Local gates the same day: make verify-api PASS, make verify-rbac PASS, go test mapping / lifecycle / translation / conformance PASS. Server dry-run of every mapped write: created. spec.vendor=acme rejected by enum.

6. How to re-run

Start the lab (this Mac is the VPN Mac)

```
scutil --nc status "F5 Access" | head -n 8
scutil --nc start "F5 Access" # if Disconnected
ping -c 2 -W 2000 172.22.1.2
```

SSH to dh2 (do not copy opi_lab_key)

```
SSH_OPTS=(-o BatchMode=yes -o ConnectTimeout=25 -o IdentityAgent=none -o IdentitiesOnly=yes -o
IdentityFile="$HOME/.ssh/opi_lab_key" -o UseKeychain=yes -o KexAlgorithms=curve25519-sha256)
ssh "${SSH_OPTS[@]}" root@172.22.1.2 'k3s --version; hostname'
```

Local gates

```
cd opi-nvidia-dpf-adapter
make verify-api && make verify-rbac
KUBEBUILDER_ASSETS="$(pwd)/bin/k8s/1.36.0-darwin-arm64" go test ./pkg/mapping/ ./pkg/lifecycle/
./pkg/translation/ ./test/conformance/ -count=1
```

Full copy-paste probes, including the live CRD field matrix and dry-runs, are in ~/opi_test_report/schema_api_lab/04_testing_commands.md. Lab connection and host table: 01_lab_connection.md. Field catalog: 02_schema_api.md.

Lab rules: use the server CPU and RAM. Do not install packages, edit /etc, systemctl, reboot, or helm-uninstall DPF. Job files only in /tmp/lab-job-<name>/ and delete afterward.